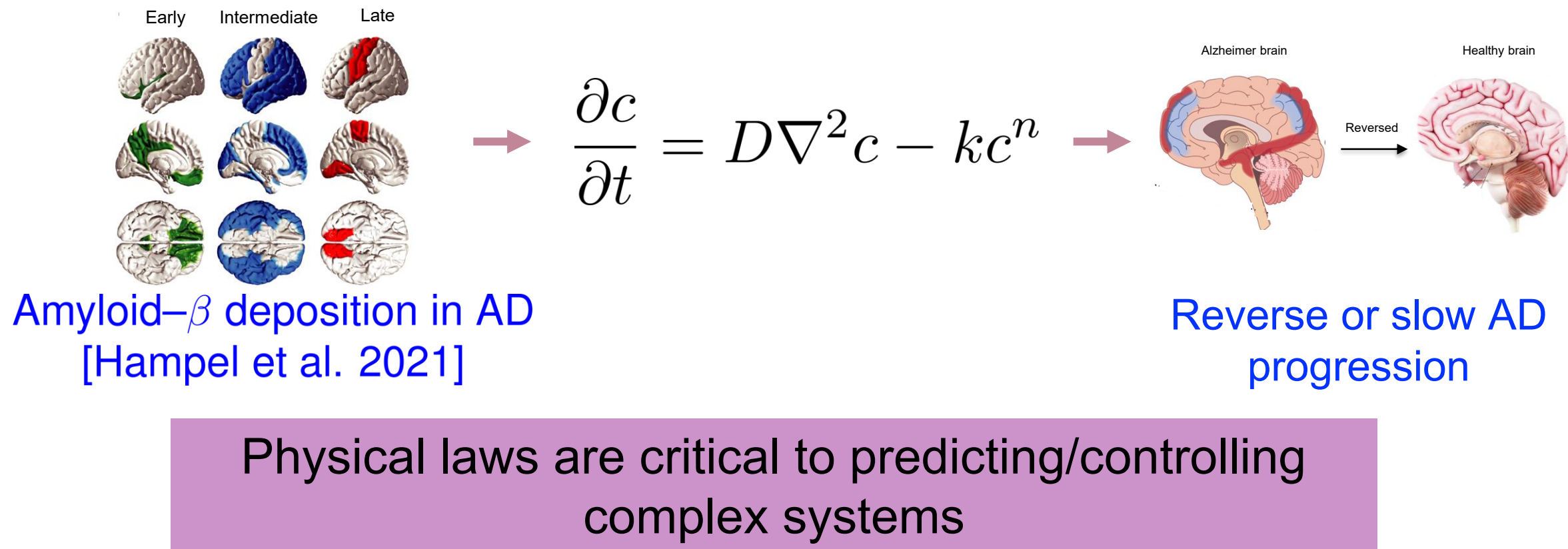
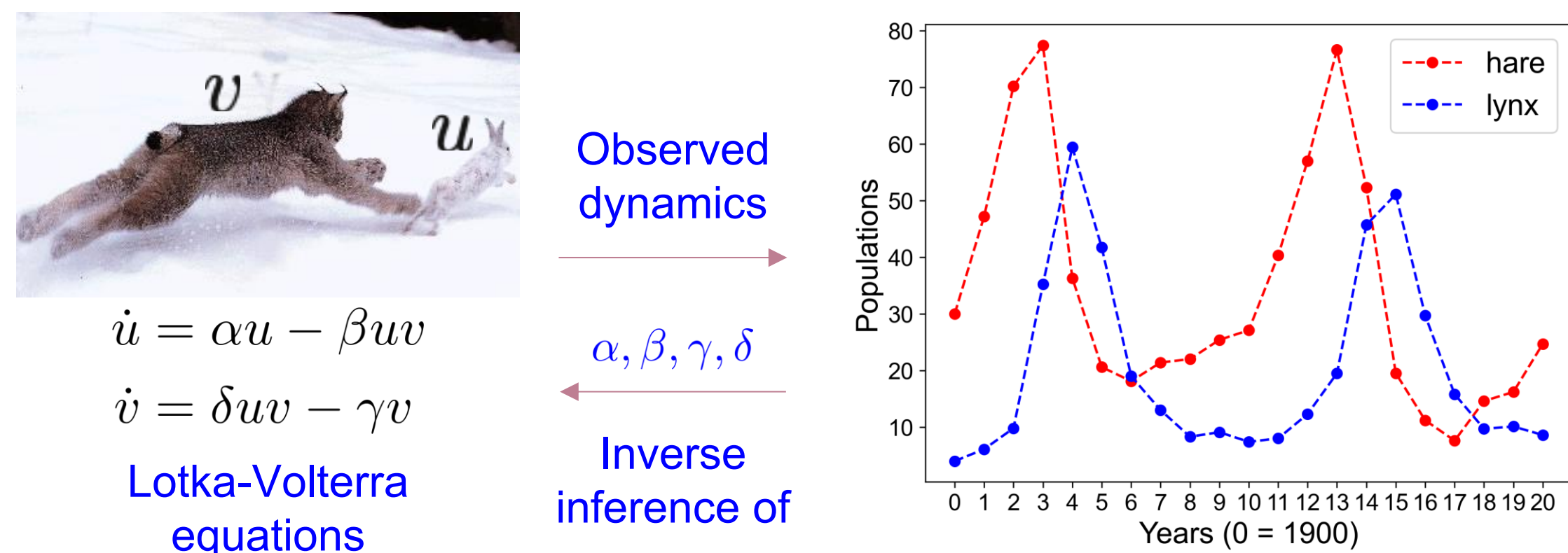


Discovering physical laws



Discovering physics from data: Example



General equation of a dynamical system

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t))$$

State variable $\mathbf{x}(t)$ Function of system variables $f(\mathbf{x}(t))$

Objective

- The **objective** of physics discovery is to infer $f(\mathbf{x}(t))$ from data $\mathbf{x}(t)$

Sparse Regression [Brunton et al.]

- Reformulate the physics using a “library” of physics

$$\frac{d}{dt}\mathbf{x}(t) = f(\mathbf{x}(t)) \rightarrow \dot{\mathbf{X}} = \Theta(\mathbf{X})\Xi$$

Library of functions $\Theta(\mathbf{X})$ Unknown coefficients Ξ

- Example of physics library for Lokta Volterra –

$$\begin{bmatrix} \dot{u} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 1 & u & v & u^2 & v^2 & uv \end{bmatrix} \begin{bmatrix} \xi_{11} & \xi_{12} \\ \xi_{21} & \xi_{22} \\ \xi_{31} & \xi_{32} \\ \xi_{41} & \xi_{42} \\ \xi_{51} & \xi_{52} \\ \xi_{61} & \xi_{62} \end{bmatrix}$$

$\Theta(u, v)$

- Physics discovery using sparse regression

$$\Xi^* = \arg \min_{\Xi} \left\| \dot{\mathbf{X}} - \Theta(\mathbf{X})\Xi \right\|_F + \Omega(\Xi)$$

Noisy measurements $\dot{\mathbf{X}}$ Regularization $\Omega(\Xi)$

Ground truth (from literature)	$\dot{u} = 0.55u - 0.45uv$ $\dot{v} = -0.84v + 0.543uv$
Discovered dynamics	$\dot{u} = -0.31 + 0.49u - 0.36uv$ $\dot{v} = 0.14 + 0.21u - 1.11v + 0.31uv + 0.16v^2$

Limitations

- Includes extra or misses key library terms
- Highly sensitive to noise and poor generalization

Bayesian Inference [Hirsh et al., 2022]:

- Quantifies uncertainty due to measurement noise
- Employs prior knowledge to regularize learning process
- Sparsity promoting prior: spike and slab

$$\xi_i | \lambda_i \sim \lambda_i \mathcal{N}(0, c^2) + (1 - \lambda_i) \mathcal{N}(0, \epsilon^2); \epsilon \ll c$$

Unknown coefficient ξ_i Sparsity parameter $\lambda_i \sim \text{Bernoulli}(\pi)$

- Likelihood: Gaussian error model

$$p(X|\Xi, x_0, \phi) = \prod_{i=1}^n \prod_{j=1}^d \left(\frac{1}{\sigma_j \sqrt{2\pi}} \right) \exp \left[-\frac{1}{2\sigma_j^2} |y_{ij} - \hat{x}_j(t_i, \Xi, x_0)|^2 \right]$$

Observed dynamic y_{ij} Predicted Dynamic $\hat{x}_j(t_i, \Xi, x_0)$

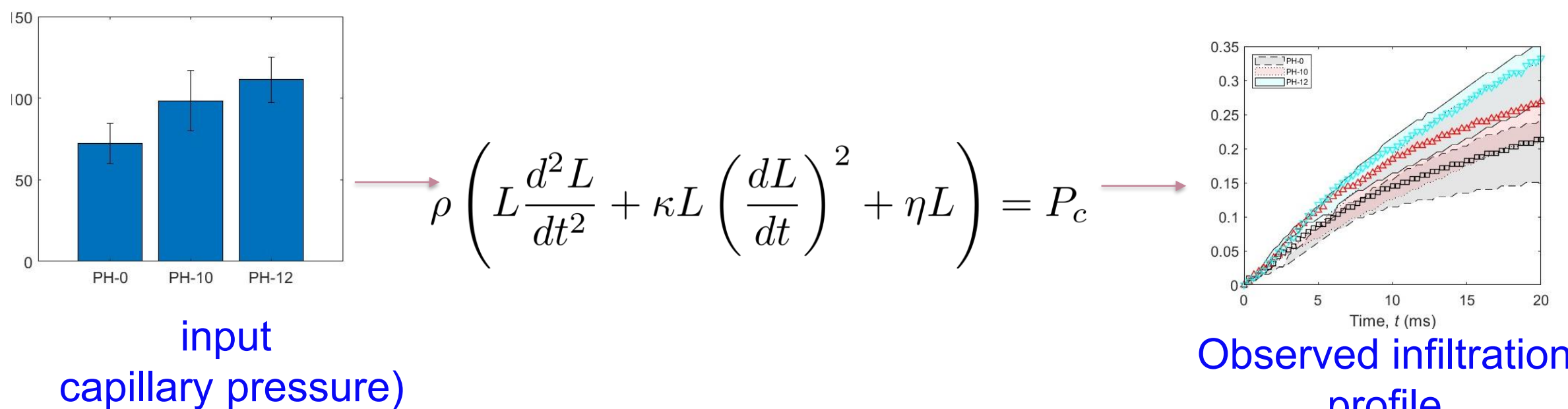
Ground truth (from literature)	$\dot{u} = 0.55u - 0.45uv$ $\dot{v} = -0.84v + 0.543uv$
Discovered dynamics	$\dot{u} = 0.75u + 0.18v - 0.54uv$ $\dot{v} = 0.14 - 0.06u - 0.98v + 0.57uv + 0.16v^2$

Limitations

- Incorrectly learns extra (higher order) terms
- Estimated coefficients are inaccurate

System Variability: Key Learning Component

- Variability in input parameters propagates to observed



- Existing methods do not account for system variability failing to recover the correct governing physics [Butler et al., 2020]

Need to account for input variability and measurement noise for robust discovery.

Stochastic Inverse Modelling

- To account for the system variability, we consider the unknown coefficients are random variables

$$\dot{\mathbf{X}} = \Theta(\mathbf{X})\underline{\lambda}$$

Library of symbolic functions $\Theta(\mathbf{X})$ Unknown (stochastic) coefficients $\underline{\lambda} \sim F(\mu_{\underline{\lambda}}, \sigma_{\underline{\lambda}}^2)$

- Accounting for measurement noise through observation

$$\mathbf{Y} = \int \dot{\mathbf{X}} dt + \epsilon = \int \Theta(\mathbf{X})\underline{\lambda} dt + \epsilon$$

Measurement noise $\epsilon \sim \mathcal{N}(0, \sigma_{\text{noise}}^2)$

- Objective: Estimate the probability distribution of $\underline{\lambda}$

How do we estimate the distribution of $\underline{\lambda}$ while accounting for measurement noise?

Pushforward Inference

- To estimate posterior distribution of $\underline{\lambda}$ ($\pi_{\underline{\lambda}}^{\text{post}}$)
 - The posterior distribution is estimated using:
- Function coefficients (random variable) $\underline{\lambda}$ Observed data $\mathbf{y}$
- Posterior density $\pi_{\underline{\lambda}}(\underline{\lambda}|\mathbf{y})$ Prior density $\pi_{\underline{\lambda}}(\underline{\lambda})$ Likelihood ratio $\frac{\pi_{\mathbf{y}}(\mathcal{Q}(\underline{\lambda}))}{\pi_{\mathcal{Q}}(\mathcal{Q}(\underline{\lambda}))}$ Observed data density $\pi_{\mathbf{y}}(\mathbf{y})$ Pushforward of prior density $\pi_{\mathcal{Q}}(\mathcal{Q}(\underline{\lambda}))$

Theoretical Results

Theorem 1- Consistency of Bootstrapped Distribution

Suppose $\mathbf{y}_1^*(t), \mathbf{y}_2^*(t), \dots, \mathbf{y}_n^*(t)$ are resampled time series from $\mathbf{y}(t)$, the observed data, and each has finite variance. Then, the bootstrapped distribution $\pi_{\mathbf{y}^*}$ is consistent and satisfies

$$K(\pi_{\mathbf{y}^*}(\mathcal{Q}(\underline{\lambda})), \pi_{\mathbf{y}}(\mathcal{Q}(\underline{\lambda}))) \xrightarrow[n \rightarrow \infty]{a.s.} 0$$

where $K(\cdot, \cdot) = D_{KL}(\pi_{\mathbf{y}} \parallel \pi_{\mathbf{y}^*}) = \int \pi_{\mathbf{y}}(x) \log \left(\frac{\pi_{\mathbf{y}}(x)}{\pi_{\mathbf{y}^*}(x)} \right) dx$ is the KL divergence.

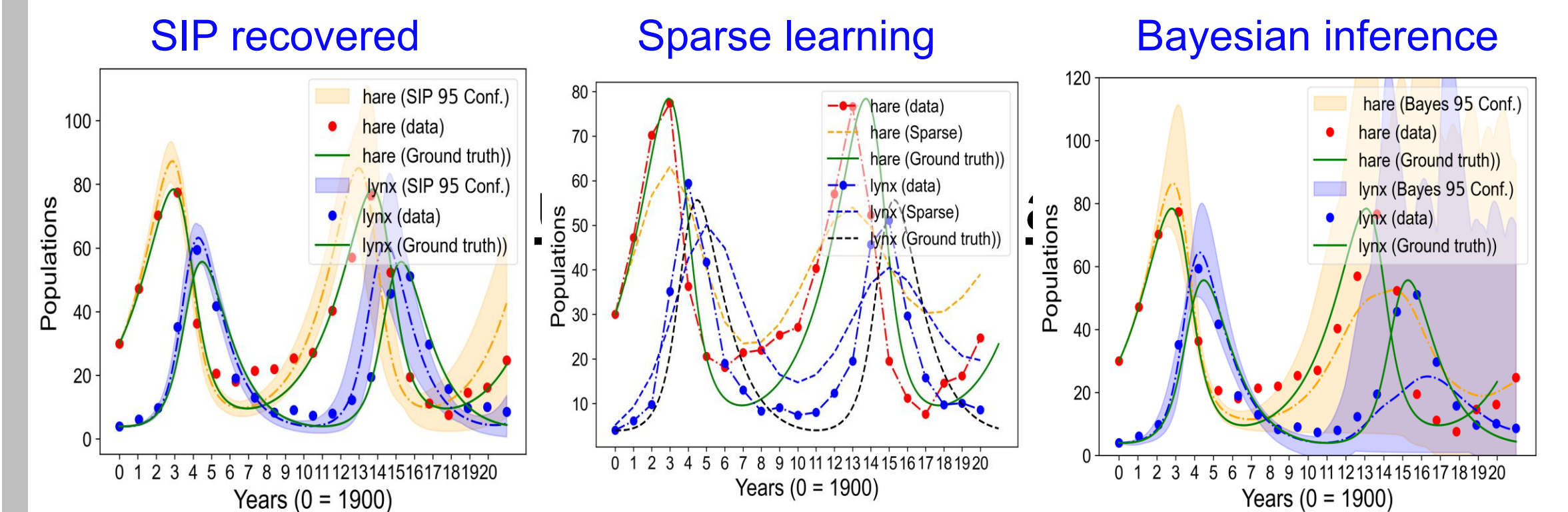
Theorem 2- Consistency of the Pushforward Method

Let $P_{\underline{\lambda}}$ be a probability measure on $(\underline{\lambda}, \Omega_{\underline{\lambda}})$. The posterior push-forward $\pi_{\underline{\lambda}}(\underline{\lambda}|\mathbf{y})$ is given by:

$$\int_{\mathbf{y}} \left(\int \pi_{\underline{\lambda}}(\underline{\lambda}) \frac{\pi_{\mathbf{y}}(\mathcal{Q}(\underline{\lambda}))}{\pi_{\mathcal{Q}}(\mathcal{Q}(\underline{\lambda}))} d\mu_{\underline{\lambda}, \mathcal{Q}(\underline{\lambda})}(\underline{\lambda}) \right) d\mu_{\mathbf{y}}(\mathcal{Q}(\underline{\lambda}))$$

This is a consistent solution to the stochastic inverse problem, i.e., the push-forward measure satisfies:

$$P_{\underline{\lambda}}(\mathcal{Q}^{-1}(A)) = P_{\mathbf{y}}^{\mathcal{Q}}(A) = P_{\mathbf{y}}(A)$$



Functions	Ground truth state		SIP state		Sparse state		Bayesian state	
	$\dot{u}$	$\dot{v}$	$\dot{u}$	$\dot{v}$	$\dot{u}$	$\dot{v}$	$\dot{u}$	$\dot{v}$
1	0	0	0	0	-0.31	0.14	0	0.14
u	0.55	0	0.6	0	0.49	0.21	0.75	-0.06
v	0	-0.84	0	-0.92	0	-1.11	0.18	-0.98
uv	-0.45	0.543	-0.45	0.54	-0.36	0.31	-0.54	0.57
v ²	0	0	0	0	0	0.16	0	0.16

Summary

- Developed a novel approach to physics discovery with 82% improved accuracy over state-of-the-art
- Accurately learned physics structure and capture system variability

Future work

- Enhance model accuracy through noise-variability separation and advanced high-dimensional sampling technique

References:

-Brunton et al., Proceedings of the National Academy of Sciences, 113. 3932-3937, 2016
-Hirsh et al, Royal Society Open Science, 9(2), 211823, 2022